INDIAN INSTITUTE OF TECHNOLOGY ROPAR

Department of Mechanical Engineering

Presentation on :

Sustainable Energy Ideas

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SOLAR MAACHINING

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ENERGY FROM MOTION OF TRAINS

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INDOOR DAYLIGHTING SYSTEM USING OPTICAL FIBRES

1. SOLAR MACHINING



IMAGE: GOOGLE-IMAGES

Design exploration with experimental limitations

CONCEPT

You may have seen concentrated solar heaters in books or TVs where concave disc is used to concentrate sunlight to a single focal point. sunlight from a wide area is concentrated onto a small absorber located at the focus, greatly increasing the energy density.

IDEA

WHY DONT WE USE THIS ENERGY TO DO MAACHINING AND WELDING

HOW

First of all transport this energy to desired location for machning annd welding

solution

we will use optical fibre

With minnimum loss

solution

we will use reflective coating on the surface of the fibres

Protect the fibres and concentrators from melting

solution

we will use recirculating waters around the fibre and coating , and also aaround the concentrator

Procedure

- Replace the usual concentrator with a high melting point glass like material designed to internally reflect and concentrate the light to its end.





side view

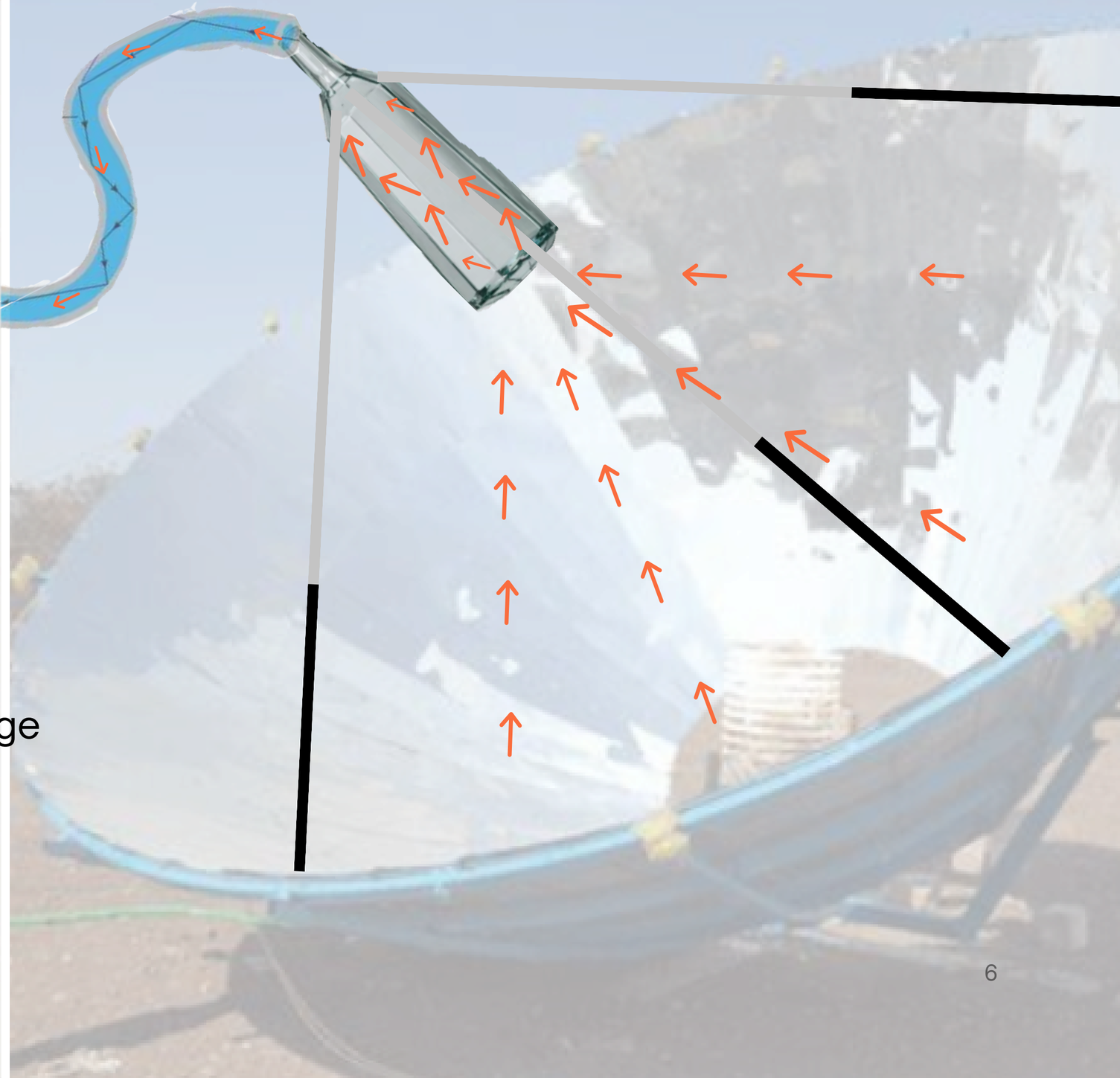
front view

IMAGE CREATED USING PROMPT

Procedure

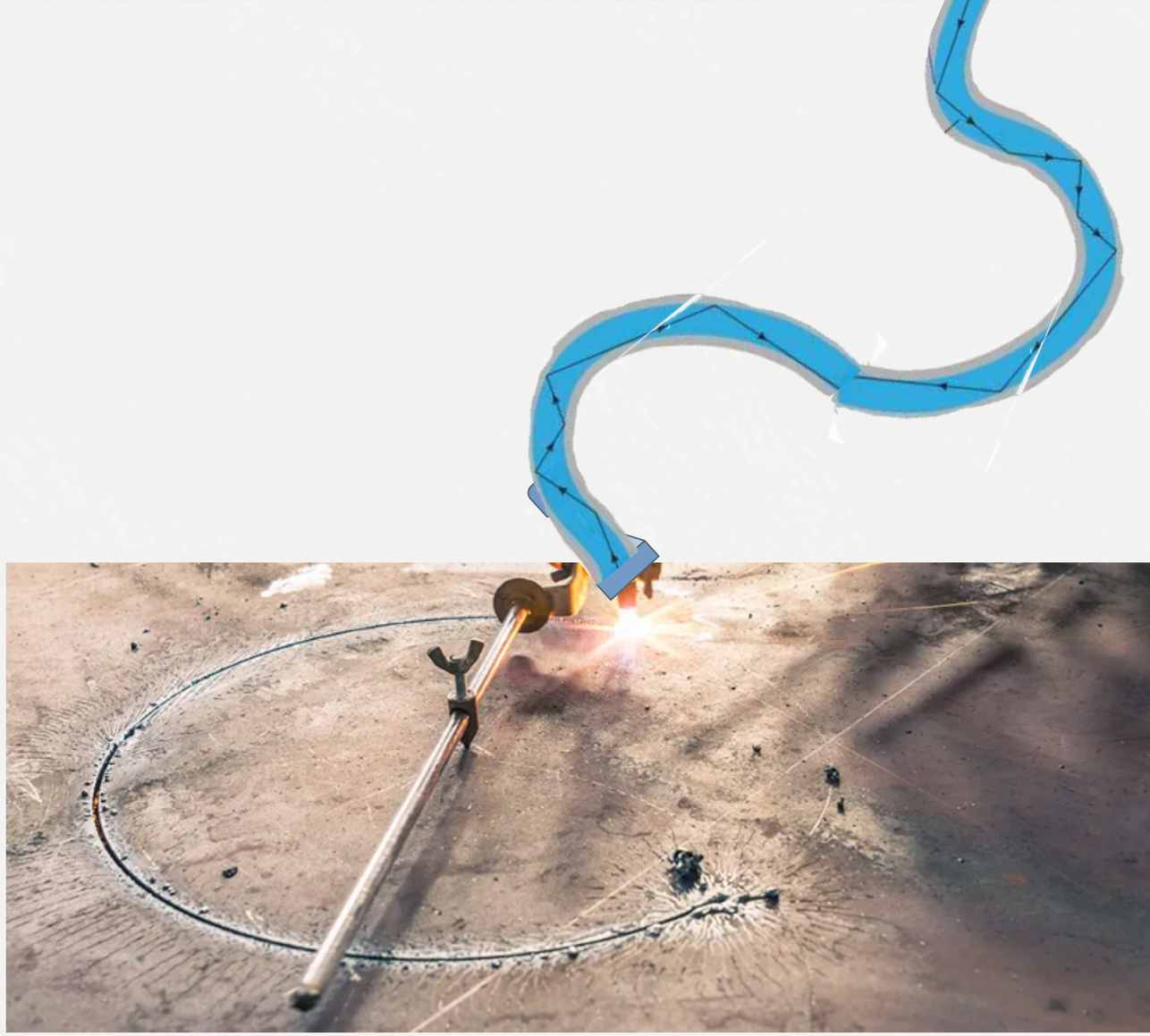
- Now the end of the glass concentrator will be connected/joined to thick optical fibres

- Joining can be done using transparent laser traansmission welding , for uninterrupted passage of light through the junction



Machining/ welding

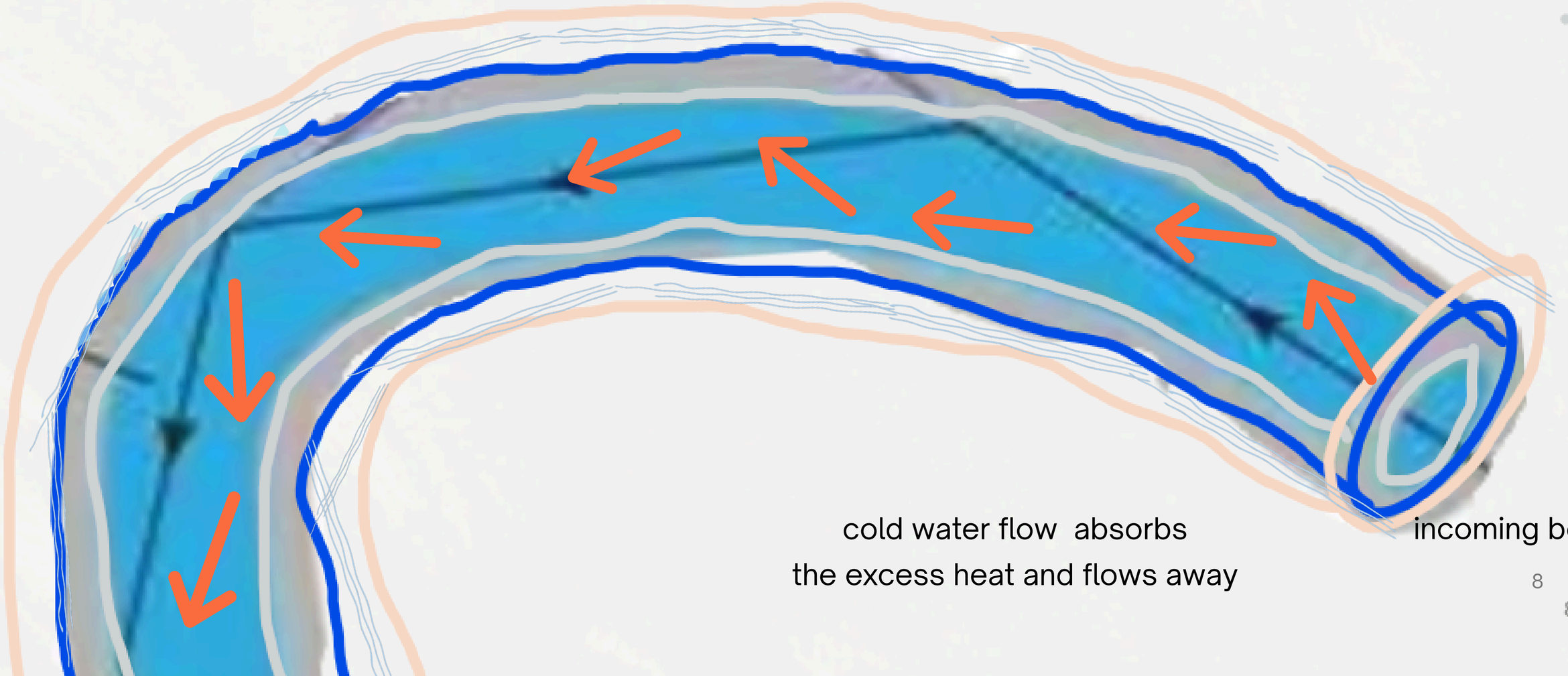
- At the other end of the optical fibre we will get hghly concentrated beam of sunlight
- The other end of the fibre caan be maade into a holding torch like those in welding and gas cutters .



melting and cuttinng thin metal sheets

Fibre coating and cooling

highly reflecting coating
made of high condctivity material
that conducts away the over heat to
the next layer of flowing water



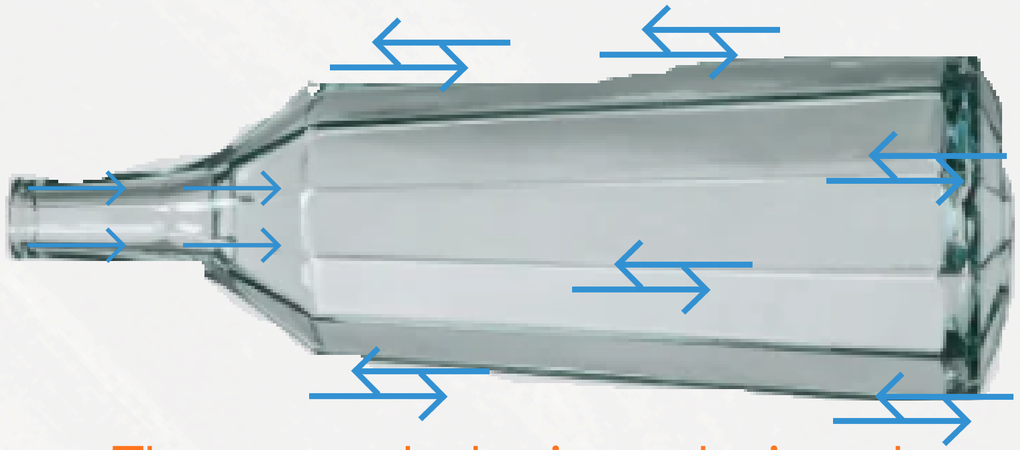
cold water flow absorbs
the excess heat and flows away

incoming beam

8

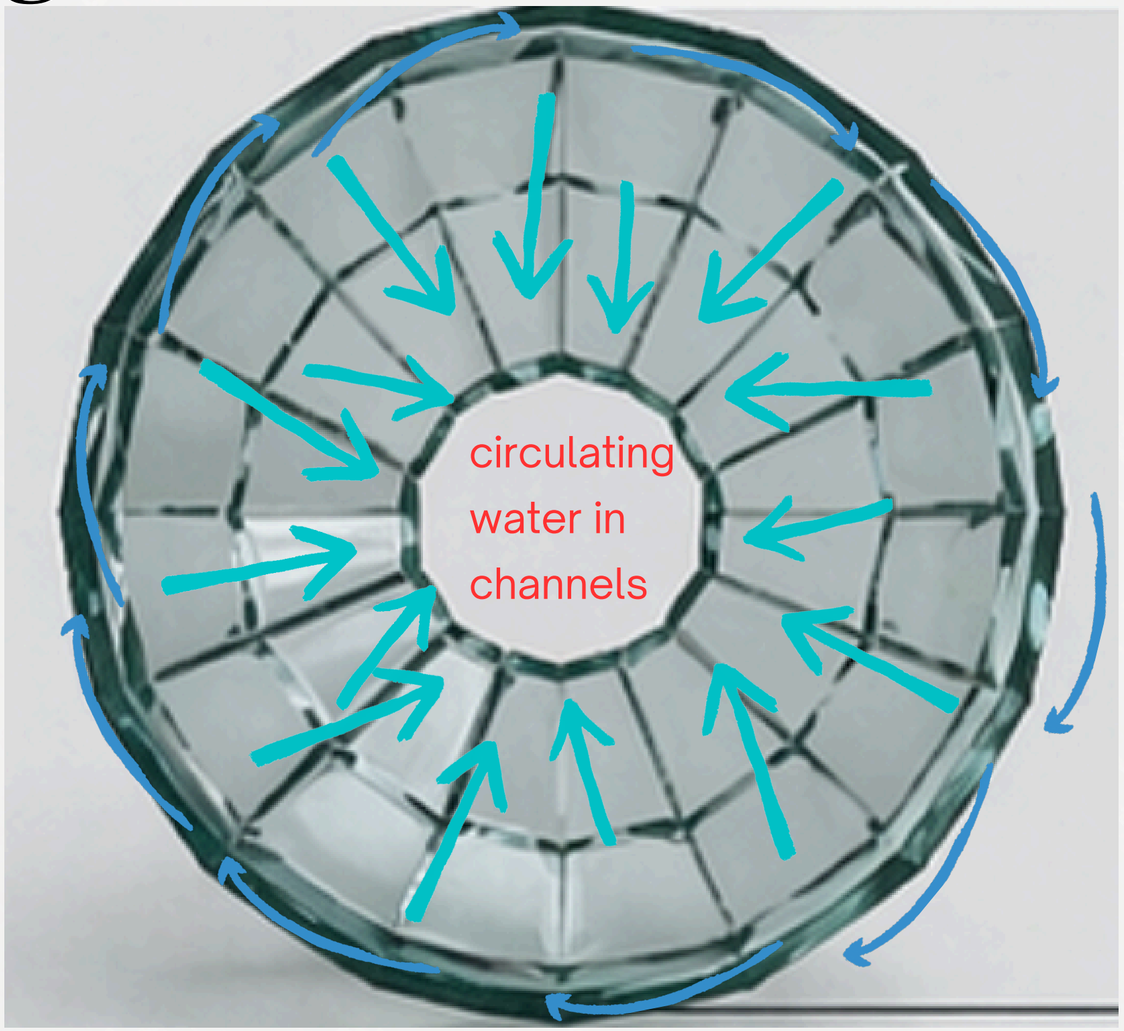
Concentrator cooling

- In a similar way , we can provide cooling to the concentrator at the focal point

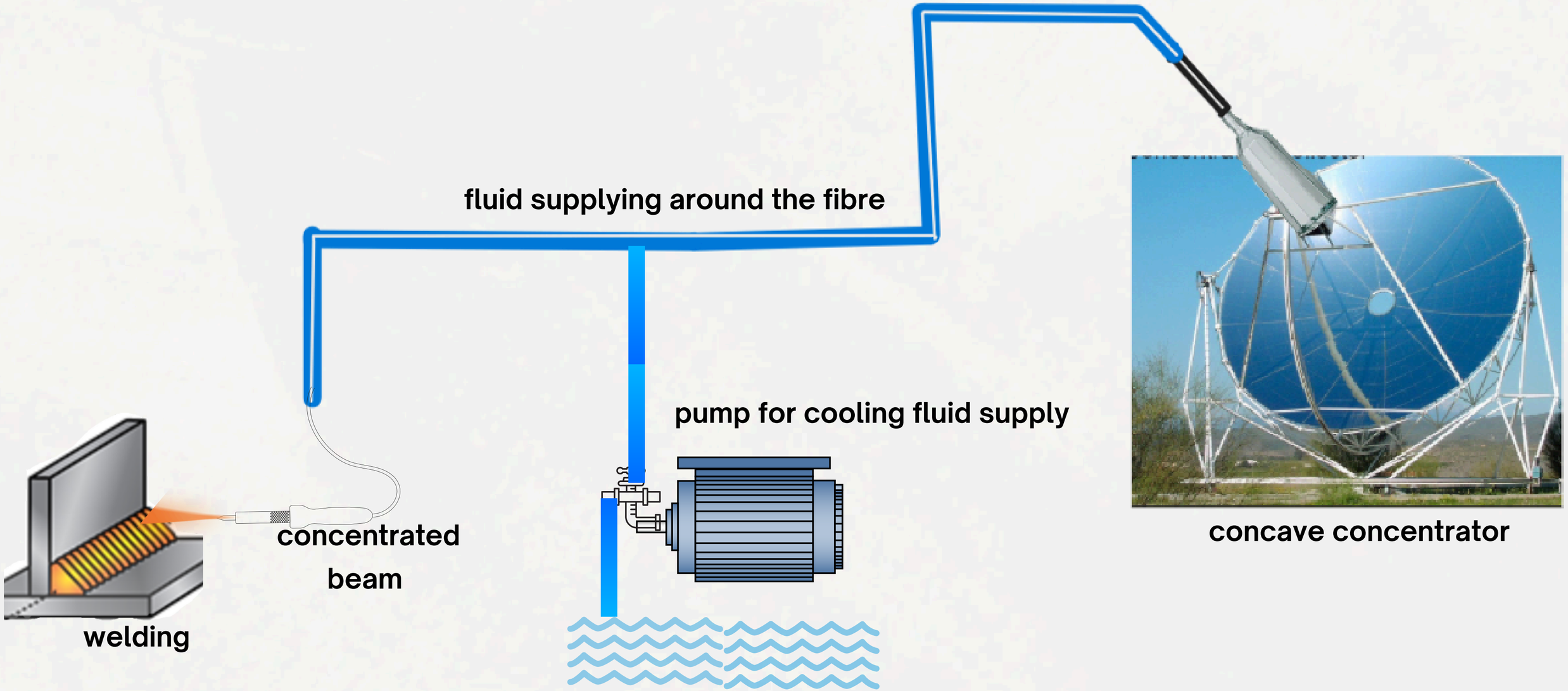


- The concentretor is so designed that it has internal micro channels for water circulation

With reflective surface coating



Fluid supply for cooling



Materials and typical calculations

Reflector area – 2 msq – 2 KWatt of sunlight

Location	Approximate Irradiance (W/m²)
Top of Atmosphere	1361
Earth's Surface (Clear Day)	~1000
Earth's Surface (Average)	~340 (global average, accounting for day/night and weather)

Source

https://earth.gsfc.nasa.gov/climate/projects/solar-irradiance/science

2 KWatt of sunlight ---- can reach 1214 deg-celc

A solar furnace is defined as a system that uses a group of curved mirrors or mirrored surfaces to concentrate sunlight into a high-temperature stream, capable of reaching temperatures up to 4000°C. This concentrated energy can be utilized for various applications, including melting metals, generating electricity, and cooking food.

Source

www.sciencedirect.com/topics/engineering/solar-furnace?

Materials and typical calculations

Concentrator material – silica (quartz glass) – melting temperature

Thermal properties of quartz glass

Deformation point: 1120℃

Softening point: 1680℃

Source

https://www.qimglass.com/Fused-Silica-Glass.html?

Coating material – Silver/copper – high conductivity + reflectivity

Thermal conductivity: 430 W/m·K (very high)
Reflectivity: 95–99% (highest for visible light) } Silver

Thermal conductivity: 400 W/m·K
Reflectivity: 90% (good, not best) } Copper

Setup capabilities

Setup parameters

Reflector power ≈ 2 kW → 2000 W

Assume focus spot ≈ 1 $cm^2 = 1 \times 10^{-4}$ m^2

Calculation of temperature for our setup

Input power=Radiation loss $P=\sigma A T^4$

Where:

P = power at focus (W)

A = focal spot area (m^2)

T = temperature (K)

σ=5.67×10−8 W/m2K4

$2000 = 5.67\times10^{-8}\times(1\times10^{-4})\times T^4$

$T4 = (2000)/(5.67\times10^{-12}) \approx 3.52\times10^{14}$

T ≈ 1360K ≈ 1087°C

Metals suitable for our setup(2 msq)

Welding		Cutting	
Al-(600 'C)	Thin sheets	Al-(600 'C)	Thin sheets
Brass-(950 'C)		Brass-(950 'C)	
Cu & mild steel of lower thickness		Cu & mild steel of lower thickness	

Applications

The system is best suited for low-to-medium power applications like thin metal processing, welding , cutting and brazing —not heavy industrial cutting (that will rrequire larger setup area).

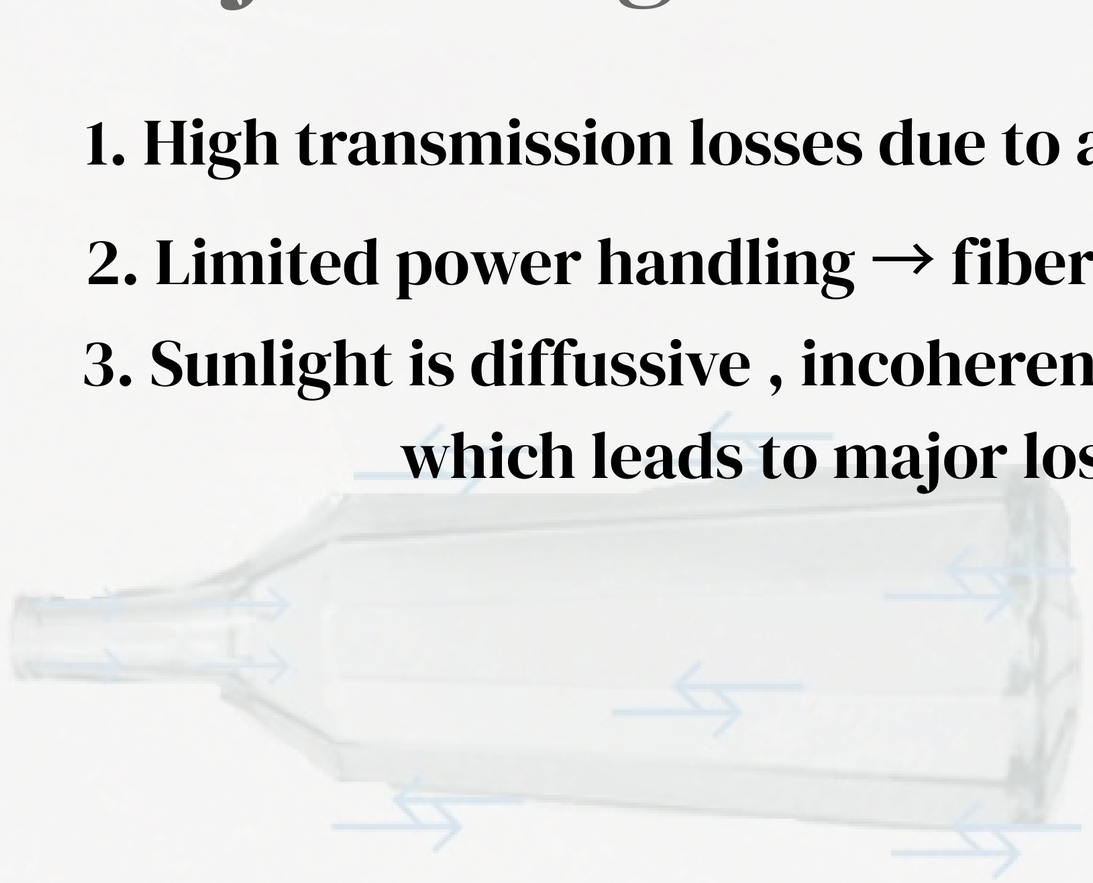
The system can be utilized as an energy provider in remote areas for small scale manufacturing operations .

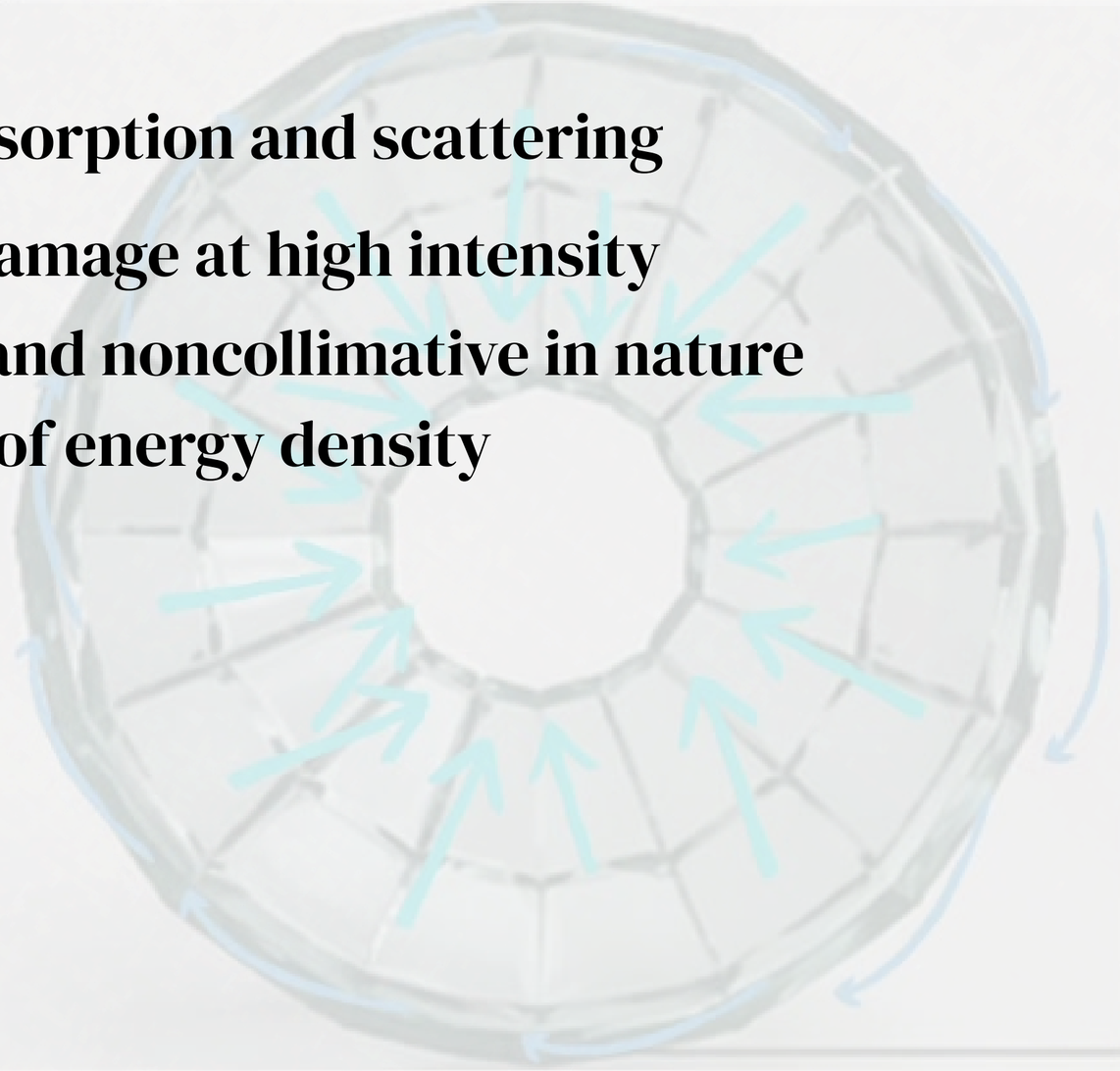




Major Design limitations

1. High transmission losses due to absorption and scattering
2. Limited power handling → fiber damage at high intensity
3. Sunlight is diffussive , incoherent and noncollimative in nature which leads to major loss of energy density





The End

THANK YOU FOR LISTENING